CBSE Class 10 Mathematics Standard

Time Allowed: 3 Hours **Maximum Marks: 70**

General Instructions:

1. This question paper contains 35 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The value of k for which the quadratic equation $2x^2 - kx + 8 = 0$ has equal real roots is

(A) (A) 4
(B) (B) 16
(C) (C) ± 8
(D) (D) ± 4

2. Which of the following equations has 2 as a root?

(A) (A) $x^2 - 4x + 4 = 0$
(B) (B) $x^2 + 3x - 12 = 0$
(C) (C) $2x^2 - 5x + 1 = 0$
(D) (D) $x^2 + 4x + 4 = 0$

3. The quadratic equation $x^2 + 5x + 6 = 0$ has roots

(A) (A) $2, 3$
(B) (B) $-2, -3$
(C) (C) $1, 6$
(D) (D) $-1, -6$

4. The discriminant of the equation $3x^2 - 2x + \frac{1}{3} = 0$ is

(A) (A) 2
(B) (B) 1
(C) (C) -1
(D) (D) 0

5. If the equation $x^2 + 4x + k = 0$ has real and distinct roots, then

(A) (A) $k > 4$

(B) (B) $k < 4$
(C) (C) $k = 4$
(D) (D) $k \leq 4$

6. The sum of the roots of the quadratic equation $x^2 - 9 = 0$ is

(A) (A) 3
(B) (B) 9
(C) (C) 0
(D) (D) -9

7. Which of these is a quadratic equation?

(A) (A) $x^2 - 3x = 0$
(B) (B) $x^3 - x^2 = 5$
(C) (C) $(x + 1)^2 = x^2 + 3$
(D) (D) $x + \frac{1}{x} = 2$

8. If $\frac{1}{2}$ is a root of the equation $x^2 - mx - \frac{5}{4} = 0$, the value of m is

(A) (A) 2
(B) (B) -2
(C) (C) 1
(D) (D) -1

9. The quadratic equation $2x^2 - \sqrt{5}x + 1 = 0$ has

(A) (A) Two distinct real roots
(B) (B) Two equal real roots
(C) (C) No real roots
(D) (D) More than two roots

10. The decimal expansion of the rational number $\frac{147}{2^2 \times 5^3 \times 7}$ will terminate after how many places of decimals?

(A) (A) 2
(B) (B) 3
(C) (C) 4
(D) (D) 5

11. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x and y are prime numbers, then the result of $\text{HCF}(a, b) \times \text{LCM}(a, b)$ is:

(A) (A) x^3y^3
(B) (B) x^4y^4

(C) (C) x^4y^5
(D) (D) x^2y^2

12. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is:

(A) (A) 2
(B) (B) 3
(C) (C) 4
(D) (D) 1

13. The largest number which divides 70 and 125 leaving remainders 5 and 8 respectively is:

(A) (A) 11
(B) (B) 17
(C) (C) 13
(D) (D) 19

14. If n is a natural number, then $9^n - 4^n$ is always divisible by:

(A) (A) 2
(B) (B) 3
(C) (C) 4
(D) (D) 5

15. The product of a non-zero rational number and an irrational number is:

(A) (A) Always rational
(B) (B) Always irrational
(C) (C) Rational or irrational
(D) (D) One

16. The total number of prime factors of 1729 is:

(A) (A) 1
(B) (B) 2
(C) (C) 3
(D) (D) 4

17. If two positive integers p and q can be expressed as $p = ab^2$ and $q = a^3b$, where a, b are prime numbers, then $\text{LCM}(p, q)$ is:

(A) (A) ab
(B) (B) a^3b^2

(C) (C) a^2b^2

(D) (D) a^3b^3

18. Which of the following rational numbers has a non-terminating repeating decimal expansion?

(A) (A) $\frac{13}{3125}$

(B) (B) $\frac{17}{8}$

(C) (C) $\frac{6}{15}$

(D) (D) $\frac{77}{210}$

19. Assertion (A): The quadratic equation $x^2 + 2kx + 9 = 0$ has real and equal roots for $k = 3$ or $k = -3$. Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has real and equal roots if and only if its discriminant $D = b^2 - 4ac$ is equal to 0.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

20. Assertion (A): The number $\frac{7}{2^3 \times 5^2}$ has a terminating decimal expansion. Reason (R): A rational number $\frac{p}{q}$ has a terminating decimal expansion if the prime factorization of q is of the form $2^n \times 5^m$, where n and m are non-negative integers.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

1. If the HCF of 306 and 657 is 9, find their LCM.

2. Without performing long division, state whether the decimal expansion of $\frac{13}{3125}$ is terminating or non-terminating repeating. If terminating, find the number of decimal places after which it terminates.

Section C: Short Answer (3 marks each)

1. A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

2. Find the value of k for which the quadratic equation $(k+1)x^2 - 2(k-1)x + 1 = 0$ has real and equal roots.

3. Solve the quadratic equation $3x^2 - 2\sqrt{6}x + 2 = 0$ by the method of completing the square or quadratic formula.

4. Show that for any positive integer n, the number $n^3 - n$ is always divisible by 6.

5. Find the greatest number that divides 108 and 192 leaving remainders 12 and 6 respectively.

6. Without performing long division, determine whether the rational number $\frac{147}{120}$ has a terminating or non-terminating repeating decimal expansion. Justify your answer.

Section D: Long Answer (4-5 marks each)

1. A train travels at a certain average speed for a distance of 540 km. If the speed of the train is increased by 9 km/h, the time taken to complete the journey is reduced by 1 hour. Form a quadratic equation in terms of the original speed x km/h, and hence find the original speed of the train.

2. Find the value of k for which the quadratic equation $(k+4)x^2 + (k+1)x + 1 = 0$ has two equal real roots. For this value of k, find the roots of the equation.

3. Show that for any positive integer n, the expression $n^3 - n$ is always divisible by 6.

4. A merchant has two types of oil: 560 litres of mustard oil and 720 litres of sunflower oil. He wants to pack these oils into tin containers of equal size such that each container is completely filled. Find the maximum capacity of such a container and the total number of containers required for both types of oil.

Section E: Case Study (4 marks each)

1. A solar panel installation firm is designing a rectangular array of solar cells on a rooftop. The length of the array is 4 m more than its width. The area of the array is 96 m^2. Let x be the width of the array in meters.

(i) Which of the following quadratic equations represents the given situation?

(A) (A) $x^2 + 4x - 96 = 0$
(B) (B) $x^2 + 4x - 96 = 0$
(C) (C) $x^2 - 4x - 96 = 0$
(D) (D) $x^2 + 4x + 96 = 0$

(ii) Find the width of the array.

(iii) What is the length of the diagonal of the array?

(A) (A) 10 m
(B) (B) 12 m
(C) (C) $4\sqrt{13}$ m

(D) (D) 15 m

(iv) If the area were instead 117 m^2 with the same relation between length and width, would the width be an integer? Justify.

2. A small community garden has a rectangular plot with a walkway of uniform width x around it. The inner garden measures 10 m by 8 m. The total area of the garden including the walkway is 168 m^2.

(i) Express the total area of the plot in terms of x.

(A) (A) $(10 + 2x)(8 + 2x)$
(B) (B) $(10 + x)(8 + x)$
(C) (C) $80 + 4x^2$
(D) (D) $10x + 8x$

(ii) Form the quadratic equation in x and simplify it to standard form.

(iii) Find the width of the walkway x.

(A) (A) 1 m
(B) (B) 2 m
(C) (C) 2.5 m
(D) (D) 3 m

(iv) If the walkway width x is doubled, what will be the new total area?

3. A packaging unit in a factory produces two types of high-quality organic fertilizers, Type-A and Type-B, in batches. The number of bags produced per batch for Type-A is x and for Type-B is y. The quality control team observes that the total number of bags x and y must satisfy the relationship $HCF(x, y) = 12$ and $LCM(x, y) = 360$. Furthermore, it is known that x and y are both multiples of 20, and $x < y$.

(i) What is the product of the number of bags x and y?

(A) (A) 3600
(B) (B) 4200
(C) (C) 4320
(D) (D) 4800

(ii) Given that x and y are multiples of 20 and $x < y$, find the possible values of x and y.

(iii) Is the number y a terminating decimal if represented as $\frac{1}{y}$?

(A) (A) No, because the denominator has prime factors other than 2 and 5.
(B) (B) Yes, because the denominator is a multiple of 2.
(C) (C) Yes, because the denominator has only 2 as a prime factor.
(D) (D) No, because the denominator is an even number.

(iv) Find the HCF of x and y using the Fundamental Theorem of Arithmetic.